

4.21-4.23 | Third-law pairs and contact forces

Assignment 5: p. 125, #21-30 all. Use $g = 9.80 \text{ m/s}^2$. Force labels identify the object exerting the force and the object receiving it.

4.21 | Sprinter and starting blocks

Question. World-class sprinters can accelerate out of the starting blocks with nearly horizontal acceleration of magnitude 15 m/s^2 . How much horizontal force must a 55 kg sprinter exert on the blocks? Which object exerts the force that propels her: the blocks or the sprinter herself?

The required forward force on the sprinter is $F = ma = (55)(15) = 825 \text{ N}$. By Newton's third law, she exerts an equal-magnitude force backward on the blocks.

She pushes the blocks backward with 825 N (about $8.3 \times 10^2 \text{ N}$). The blocks exert the forward force that propels her.

4.22 | Small car pushing a large truck

Question. A 380 kg car pushes a 900 kg truck due east on a level road. The car exerts a horizontal force of 1600 N on the truck. What is the magnitude of the force the truck exerts on the car?

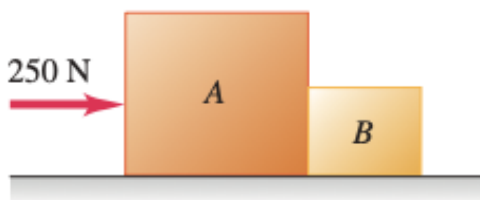
These contact forces are a third-law pair. Their magnitudes are equal even though the vehicles have different masses.

The truck exerts 1600 N on the car, directed west.

4.23 | Two boxes pushed together

Question. Boxes A and B touch on a horizontal, frictionless surface (Fig. E4.23). A has mass 20.0 kg ; B has mass 5.0 kg . A horizontal force of 250 N acts on A. Find the magnitude of the force A exerts on B.

Figure E4.23



First treat both boxes as one system:

$$a = F/(m_A + m_B) = 250/(20.0 + 5.0) = 10.0 \text{ m/s}^2.$$

Then isolate B. Its only horizontal force is the contact force from A:

$$F_{A \text{ on } B} = m_B a = (5.0)(10.0) = 50 \text{ N}.$$

A exerts 50 N on B, directed right.

4.24-4.25 | Reaction forces and acceleration

4.24 | Elevator passenger

Question. The floor exerts an upward normal force of 620 N on an elevator passenger who weighs 650 N. What are the reaction forces to these two forces? Is the passenger accelerating? If so, find the magnitude and direction of the acceleration.

Third-law partners:

- Floor on passenger: 620 N upward. Its partner is **passenger on floor: 620 N downward**.
- Earth on passenger: 650 N downward. Its partner is **passenger's gravitational pull on Earth: 650 N toward the passenger** (upward locally).

The normal force and weight are **not** a third-law pair: both act on the passenger. Their actual third-law partners act on other objects.

Acceleration: Choose upward as positive.

$$m = W/g = 650/9.80 = 66.3265 \text{ kg.}$$

$$F_{\text{net},y} = N - W = 620 - 650 = -30 \text{ N.}$$

$$a_y = -30/66.3265 = -0.452308 \text{ m/s}^2.$$

The passenger accelerates downward at 0.452 m/s².

Downward acceleration does not by itself tell you whether the passenger is moving up or down.

4.25 | Earth accelerates toward a falling student

Question. A 45 kg student jumps from a high diving board. What is Earth's acceleration toward her as she accelerates toward Earth at 9.8 m/s²? Use Earth's mass 6.0×10^{24} kg, and assume the net force on Earth is the gravitational force she exerts on it.

Earth pulls the student with magnitude $F = mg = (45)(9.8) = 441 \text{ N}$. She pulls Earth with the same force magnitude, by Newton's third law.

$$a_{\text{Earth}} = F/M_{\text{Earth}}$$

$$a_{\text{Earth}} = 441/(6.0 \times 10^{24}) = 7.35 \times 10^{-23} \text{ m/s}^2.$$

Earth's acceleration is approximately $7.4 \times 10^{-23} \text{ m/s}^2$ toward the student.

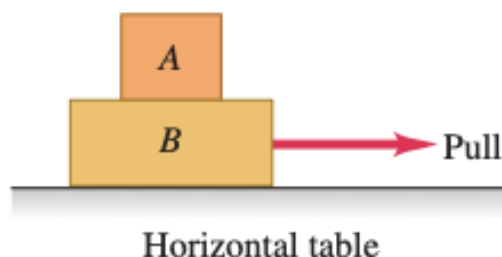
The forces have equal magnitudes; the accelerations differ because $a = F/m$ and Earth has a much larger mass.

4.26 | Free-body diagrams for the upper block

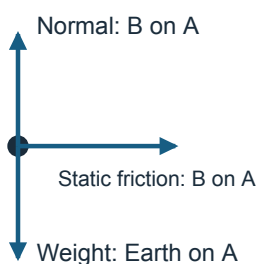
4.26 | Block A rides on block B

Question. You pull horizontally on block B in Fig. E4.26, causing both blocks to move together as a unit. Draw a carefully labeled free-body diagram of A if (a) the table is frictionless and (b) there is friction between B and the table, with the pull equal in magnitude to the table's friction force on B.

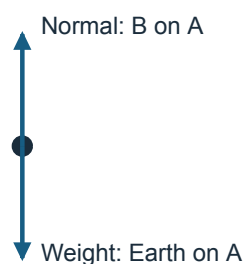
Figure E4.26



(a) A accelerates right



(b) A moves at constant velocity



(a) Frictionless table: The pull accelerates the two-block system to the right. A has no directly applied pulling force; **static friction from B** is the force that accelerates A. Friction points right to prevent A from slipping backward relative to B.

For A:

Vertical: $N_{B \text{ on } A} - m_A g = 0$.

Horizontal: $f_{s, B \text{ on } A} = m_A a$, directed right.

(a) Draw three forces on A: normal up, weight down, and static friction right.

(b) Pull equals table friction: For the combined system, the net horizontal external force is zero. Thus $a = 0$ and both blocks move at constant velocity. For A, any nonzero horizontal friction would cause acceleration, so its actual friction force is zero.

(b) Draw only normal up and weight down on A; they are equal. Friction on A is zero.

Keep the surfaces separate: Friction between B and the table acts on B, not directly on A. A rough A-B interface can have zero static friction when no friction force is needed.

4.27 | Two crates: forces on each object

4.27 | Contact forces and motion

Question. Crates A and B rest side by side on a frictionless horizontal surface, with masses m_A and m_B . A horizontal force F is applied to A and the crates move right. (a) Draw labeled free-body diagrams for A and B. Identify any third-law action-reaction pairs. (b) If F is smaller than their total weight, will the crates move? Explain.

Let C be the magnitude of the contact force between the crates. A is on the left and pushes B to the right.



(a) Equations for each crate:

$$A: F - C = m_A a; \quad N_A = m_A g.$$

$$B: C = m_B a; \quad N_B = m_B g.$$

Adding the horizontal equations cancels the internal contact forces:

$$a = F/(m_A + m_B).$$

The contact-force magnitude is $C = m_B F/(m_A + m_B)$.

The third-law pair shown is A's force on B (right) and B's force on A (left).

Each weight's reaction acts on Earth; each normal force's reaction acts on the floor. Those partners are outside these two diagrams. A crate's normal force and weight are not third-law partners.

(b) Yes. Weight acts vertically and is balanced by the floor's normal force. Because the surface is frictionless, it supplies no opposing horizontal friction.

Any nonzero horizontal F produces $a = F/(m_A + m_B) > 0$, even if $F < (m_A + m_B)g$.

4.28-4.29 | Stopping forces and an accelerating train

4.28 | Bullet stopping in a tree

Question. A .22-caliber rifle bullet traveling at 350 m/s penetrates a tree to a depth of 0.130 m. Its mass is 1.80 g. Assume a constant retarding force. (a) How long does the bullet take to stop? (b) What force, in newtons, does the tree exert on it?

Choose the initial motion as positive; $m = 0.00180$ kg. From $v_f^2 = v_i^2 + 2a\Delta x$:

$$a = (0 - 350^2)/(2 \times 0.130) = -4.71154 \times 10^5 \text{ m/s}^2.$$

$$(a) t = (v_f - v_i)/a = (-350)/(-4.71154 \times 10^5) = 7.42857 \times 10^{-4} \text{ s}.$$

$$(b) F = ma = (0.00180)(-4.71154 \times 10^5) = -848.077 \text{ N}.$$

(a) $7.43 \times 10^{-4} \text{ s} = 0.743 \text{ ms}$.

(b) 848 N opposite the bullet's motion.

The retarding force dominates during this brief stop; gravity is negligible in the stopping calculation.

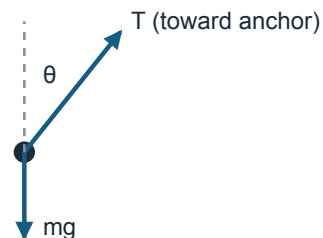
4.29 | Ball hanging in an eastbound train

Question. A ball hangs from a long string tied to a train's ceiling. The train travels east on horizontal tracks, and the ball is motionless relative to the train. Draw a labeled free-body diagram if (a) the train has uniform velocity and (b) the train speeds up uniformly. Is the net force zero in either case? Explain.

(a) Constant train velocity



(b) Train accelerates east



East is right. Only real forces are shown.

(a) Uniform velocity: The ball has zero acceleration. The string hangs vertically; tension and weight balance: $T = mg$, $F_{\text{net}} = 0$.

(b) Speeding up east: Although stationary relative to the train, the ball accelerates east relative to the ground. It hangs **west of its attachment point**, so tension points up-east. The horizontal tension component accelerates the ball.

With θ measured from vertical:

$$T \cos \theta = mg; \quad T \sin \theta = ma.$$

Thus $\tan \theta = a/g$. The net force is **ma east, not zero**.

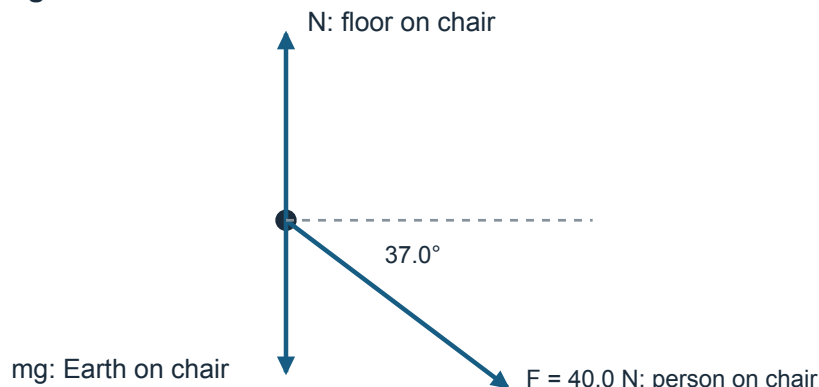
Use the ground as an inertial frame. Draw only tension and gravity; acceleration is not an additional force.

4.30 | A downward push increases the normal force

4.30 | Chair on a frictionless floor

Question. A 12.0 kg chair sits on a frictionless horizontal floor. You push on it with a force of 40.0 N directed 37.0° below horizontal, and it slides along the floor. (a) Draw a carefully labeled free-body diagram of the chair. (b) Use the diagram and Newton's laws to calculate the normal force exerted by the floor.

Free-body diagram of the chair



Frictionless floor: no horizontal friction force.

(a) The three forces:

- N: upward normal force from floor on chair.
- mg: downward gravitational force from Earth on chair.
- F: person's applied force on chair, 37.0° below horizontal.

There is no friction force. Resolve F into components when writing equations; the components are not extra forces to add alongside F.

(b) Vertical force balance: The chair stays on the floor, so $a_y = 0$, even while it accelerates horizontally.

$$\Sigma F_y = N - mg - F \sin 37.0^\circ = 0.$$

$$\mathbf{N = mg + F \sin 37.0^\circ.}$$

$$N = (12.0)(9.80) + (40.0) \sin 37.0^\circ$$

$$N = 117.6 + 24.0726 = 141.673 \text{ N.}$$

Normal force: N = 142 N upward.

Check: $N > mg$ because the floor supports the chair's weight plus the downward component of your push.